

Executable LLM2FPGA Survey Evidence Report

1 Question, scope, and amendment

The frozen protocol reconciles 459 to 461 source records without changing the study question or eligibility rules. Evidence: `survey/protocol.md#lines=5-17`, `survey/config/scope.yaml`, `LLM-inference-on-FPGA-papers/data/catalog.json`, and `survey/build/provenance.json`.

2 Methods and corpus

Every record is retained in `survey/build/records_normalized.csv`; duplicate lineage is in `survey/build/duplicate_groups.csv`; and controlled decisions are in `survey/data/screening_decisions.csv`. The final corpus is 461 source records, 456 unique works, and 5 duplicate manifestations. Final counts are A=30, B=57, C=75, D=68, X=231, with 230 included records and 231 exclusions; see `survey/build/final_flow_counts.json`.

3 Review and audit

The final grouping has 226 project families and the bounded review has 36 reviewed project-family/control rows. Evidence: `survey/build/project_families.csv`, `survey/build/deep_review.csv`, `survey/build/artifact_inventory.csv`, and `survey/build/repository_audit.csv`.

4 Compatibility and MLIR/CIRCT

R1–R8 receipts are validated before reporting. R1 stops at RTL generation rather than passing RTL; see `survey/compatibility/R1-mlir-circt/manifest.json`, `survey/compatibility/R1-mlir-circt/commands.sh`, and `survey/compatibility/R1-mlir-circt/stderr.log`. The source-faithful MLIR/CIRCT catalog is 3 projects x 20 canonical transformations = 60 assessed cells; see `survey/build/mlir_circt_stage_matrix.csv`. H1 — unsupported for the current route. H2 — bounded/partial support only. H3 — inconclusive. Evidence: `survey/build/mlir_circt_stage_matrix.md#lines=90-94`.

5 Selection result and limits

The decision matrix at `survey/build/decision_matrix_scored.csv` yields `NO_PRIMARY_ROUTE_PASSED`. Primary route: none. R1 is the ineligible next evidence-gathering route; R2 is an ineligible different-family hypothesis. The route selection rationale is `survey/build/route_selection.md`. The target board remains unspecified, and no route is claimed as an eligible end-to-end causal-LM implementation.

6 Reproduction

Run `nix develop -c python -m unittest discover -s tests -p 'test_survey_*.py'`
-v, then `nix develop -c python survey/scripts/build_report.py -root . -out`
`survey/build`. The pinned `flake.nix` and `flake.lock` development shell supplies the de-
clared `pdflatex` command rather than a host-profile TeX installation. The renderer validates every
evidence path, renders Mermaid source, invokes `pdflatex` twice, and records the transcript and
PDF SHA-256.